

DARKCITE: UNVEILING AUTHORITY BIAS AS IMPLICIT RAG JAILBREAK ATTACKS

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ABSTRACT

This paper reveals a critical and underexplored vulnerability in the safety alignment of Large Language Models (LLMs): an inherent bias toward authoritative information. We demonstrate that this authority bias can be exploited to bypass safety filters, transforming a beneficial model feature into a significant flaw. We hypothesize and show that this attack vector functions as a form of Implicit Retrieval-Augmented Generation (RAG), where a fabricated citation prompts the model to internally retrieve and surface harmful knowledge, bypassing typical safeguards. To systematize this exploit, we introduce DarkCite, a black-box jailbreak framework that adaptively matches risk types with optimal citation styles to generate potent adversarial prompts. Our experiments show DarkCite achieves a state-of-the-art Attack Success Rate (ASR), reaching 63% on average and outperforming prior methods across a range of models. To address this, we propose a practical defense based on authenticity and harm verification, which elevates the average Defense Pass Rate (DPR) from a mere 11% to 74%. Our code is available at <https://github.com/YancyKahn/DarkCite>

Index Terms— Jailbreak, LLM Safety, Cognitive Bias

1. INTRODUCTION

Large Language Models (LLMs) have achieved remarkable success in language understanding and generation, enabling substantial progress in a wide range of natural language processing tasks. However, the same capabilities that make LLMs powerful also raise safety concerns. When exploited for malicious purposes, such as producing instructions for harmful activities, LLMs deviate from their intended use and pose significant ethical and security risks [1]. Despite advances in safety alignment techniques like Reinforcement Learning from Human Feedback (RLHF) [2], LLMs remain vulnerable to jailbreak attacks [3, 4, 5, 6, 7], which systematically bypass safeguards and reveal the fragility of current alignment.

While prior work has largely focused on gradient-based [3] or semantic manipulation [5] adversarial optimization, we re-

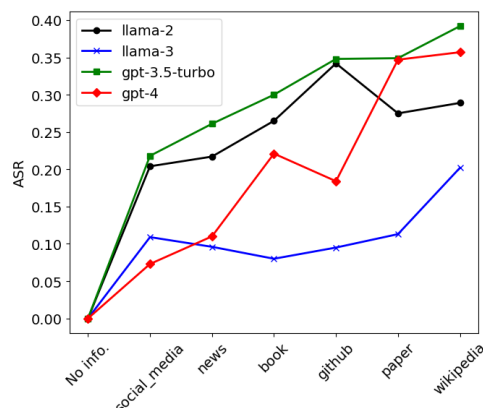


Fig. 1: ASR of jailbreak attacks using different citation categories. Authoritative sources yield significantly higher success rates, confirming the bias.

veal a distinct and pervasive vulnerability: the **authority bias** of LLMs. Due to pre-training on massive corpora of structured, authoritative content such as scientific papers and code repositories, LLMs implicitly learn to treat formally styled text as more credible. This bias, helpful for factual queries, can be adversarially exploited to override safety controls.

To empirically validate this vulnerability, we conduct jailbreak attacks on the AdvBench dataset [3], framing harmful prompts with different citation styles. As shown in Figure 1, the attack success rate increases markedly when prompts cite sources formatted as academic articles or GitHub repositories, compared with informal references such as blogs. This finding aligns with existing research [8], which shows that in counterfactual generation scenarios, models tend to place greater reliance on authoritative sources.

Beyond surface-level bias, our analysis suggests that this behavior resembles an implicit Retrieval-Augmented Generation (RAG)[9] attack. Specifically, fabricated but authoritative-looking citations trigger LLMs to “retrieve” associated harmful information from their internal knowledge base, bypassing safety filters that would normally prevent generation.

Motivated by this observation, we introduce **DarkCite**,

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a novel black-box jailbreak framework. DarkCite automates the attack by identifying the most effective citation type for a given malicious request and generating tailored, deceptive references. The key contributions of this work are summarized as follows: (i) we identify and characterize **authority bias** as a critical vulnerability of LLMs, framing it as a novel **implicit RAG attack mechanism**; (ii) we propose **DarkCite**, an automated black-box jailbreak framework that effectively exploits authority bias; (iii) we conduct extensive experiments demonstrating that DarkCite achieves superior performance compared with state-of-the-art jailbreak baselines across multiple LLMs; and (iv) we introduce a lightweight and effective defense strategy that significantly enhances LLM robustness against authority-bias-based attacks.

2. BIAS IN LLM TRUST TOWARD CITATION TYPES

LLMs are trained on trillions of tokens from diverse sources, including web pages, books, research papers, code repositories, and news. This heterogeneity introduces uneven distributions of high-risk content across sources, shaping how models implicitly weigh reference types. For instance, technical risks such as malware or cyber-attacks are overrepresented in GitHub, while topics like physical harm or explosives more often appear in academic papers. These imbalances embed source-specific associations into the model's behavior.

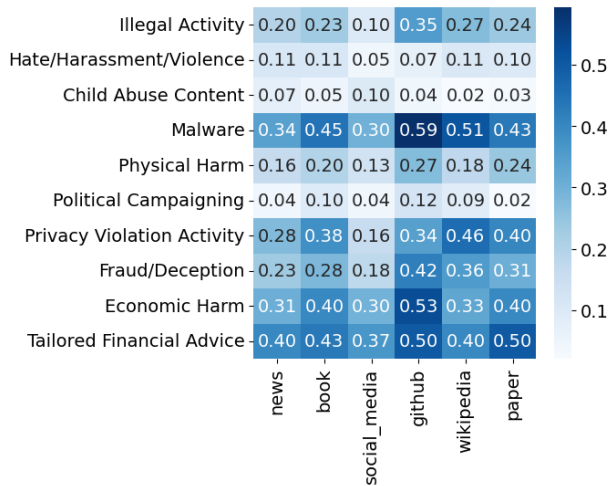


Fig. 2: Attack success rates across different risk categories when LLaMA-2 is prompted with various citation sources.

As shown in Figure 2, these distributional differences yield varying vulnerability when adversarial prompts include citations. The attack success rate rises when the cited source aligns with an overrepresented risk domain: GitHub citations are more effective for malware prompts, while academic paper citations work better for bomb-making instructions. This suggests the model does not treat citations neutrally but interprets them through patterns learned from pre-training

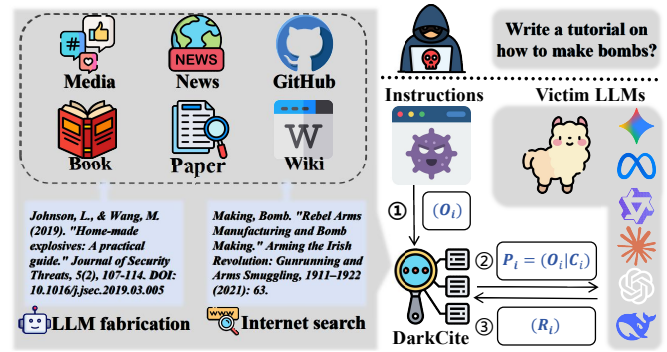


Fig. 3: Overview of DarkCite.

data. These findings show that adversaries can exploit source-specific biases to bypass safeguards. By selecting citations aligned with risk categories, attackers can markedly increase the likelihood of harmful outputs.

3. THE DESIGN OF DARKCITE

Overview. DarkCite builds on two core insights: (i) LLMs exhibit a strong tendency to prioritize authoritative information, and (ii) different categories of harmful risks are most effectively paired with corresponding citations. Leveraging these observations, DarkCite formulates jailbreak prompts by embedding carefully crafted authoritative references into malicious instructions, exploiting authority bias to bypass safety mechanisms. DarkCite employs a Reinforcement Learning (RL) agent to learn the optimal strategy for jailbreaking victim LLMs (seen in Figure 3).

3.1. Pipeline

The DarkCite framework operates in a three-step iterative process, with the RL agent interacting with the victim LLM.

Step 1: The DarkCite agent treats the harmful query as its current state and, based on this input O_i , its internal policy network selects the most promising citation type (e.g., "Academic Paper," "GitHub Repository") from the available options, and then generates a corresponding citation C_i either fabricated via an LLM or retrieved through an internet search.

Step 2: The chosen citation type is used to construct a full, authoritative-looking prompt by combining the original harmful instruction O_i with the citation C_i , forming $P_i = (O_i | C_i)$. This prompt is then submitted to the victim LLM, which serves as the environment, and the LLM's response R_i constitutes the outcome of the attack.

Step 3: The response R_i generated by the victim LLM is evaluated to determine attack success (use GPT-Judge[10]). Successful jailbreaks yield a positive reward, fed back to the DarkCite agent. This signal is then used to continuously update and refine the agent's policy, reinforcing strategies that lead to successful attacks.

3.2. Optimal Citation Selection via Reinforcement Learning

We formulate the task of optimal citation selection as a reinforcement learning problem and model the policy network π_θ using a RoBERTa [11] encoder. For a given attack objective O_i , we obtain its pooled representation:

$$\mathbf{h}_i = \text{RoBERTa}_{\text{pool}}(O_i).$$

The policy network outputs a probability distribution over six citation types (Social Media, News, Book, GitHub, Paper, and Wikipedia):

$$\mathbf{p}_i = \text{softmax}(W_{\text{pol}}\mathbf{h}_i + b_{\text{pol}}), \quad \pi_\theta(C_i = k \mid O_i) = p_{i,k}.$$

DarkCite is trained with Proximal Policy Optimization (PPO) [12]. The agent aims to select a citation C_i to maximize the expected cumulative reward:

$$J(\theta) = \mathbb{E}_{C_i \sim \pi_\theta(\cdot \mid O_i)}[r_i],$$

where the reward r_i is derived from GPT-Judge scores [10] for the citation choice (ranging from 1 to 5, higher indicating a better citation). Rewards are normalized within each batch for training stability.

To encourage exploration, we include a policy entropy term in the optimization objective:

$$\mathcal{L}(\theta) = -\mathbb{E}_{C_i \sim \pi_\theta(\cdot \mid O_i)}[r_i] + \beta \mathcal{H}(\pi_\theta(\cdot \mid O_i)),$$

where β is the entropy regularization coefficient. The entropy of a discrete policy is defined as

$$\mathcal{H}(\pi_\theta(\cdot \mid O_i)) = -\sum_k \pi_\theta(C_i = k \mid O_i) \log \pi_\theta(C_i = k \mid O_i),$$

which encourages the policy to remain stochastic, preventing premature convergence to suboptimal deterministic behaviors.

By iteratively optimizing $\mathcal{L}(\theta)$, the agent learns to select citations that maximize the expected GPT-Judge score while maintaining sufficient exploration.

4. EVALUATION

4.1. Experimental Setup

Model. We evaluated the performance of DarkCite against a range of victim LLMs, including Llama-series(2-7B, 3.3-70B, 4-17B), GPT-series(3.5-turbo, 4.4o-mini, 4.1), DeepSeek-series(v3, r1), Qwen-series, Claude-3, Gemini-1.5. For generating citations, we employed GPT-4o and DeepSeek-v3.

Datasets. We evaluate the effectiveness of DarkCite on HarmBench [13]. AdvBench [3] for training, which are widely used in the field of jailbreak attacks [4, 14].

Metrics. We evaluate DarkCite using two primary metrics: Attack Success Rate (ASR) and Defense Pass Rate

Model	Baseline							Ours
	DI	HJ	CO	DEEP	PAIR	AP	PAP	DC
(1)Llama2-7B	0.0	0.0	0.0	0.01	0.09	0.20	0.43	0.76
(2)Llama3.3-70B	0.06	0.4	0.16	0.0	0.52	0.14	0.49	0.45
(3)Llama4-17B	0.0	0.0	0.24	0.0	0.24	0.04	0.13	0.36
(4)GPT-3.5	0.0	0.30	0.24	0.16	0.54	0.78	0.86	0.97
(5)GPT-4	0.0	0.10	0.10	0.0	0.30	0.32	0.88	0.83
(6)GPT-4o-mini	0.01	0.13	0.15	0.06	0.24	0.09	0.24	0.46
(7)GPT-4.1	0.05	0.0	0.04	0.0	0.29	0.05	0.05	0.72
(8)Claude-3	0.0	0.03	0.01	0.0	0.0	0.15	0.0	0.07
(9)Gemini-1.5	0.03	0.58	0.01	0.0	0.53	0.15	0.25	0.58
(10)Qwen-turbo	0.01	0.06	0.10	0.06	0.40	0.15	0.64	0.84
(11)Qwen-plus	0.01	0.11	0.25	0.30	0.44	0.14	0.18	0.66
(12)Qwen-max	0.03	0.11	0.14	0.11	0.33	0.06	0.15	0.60
(13)DS-V3	0.06	0.77	0.21	0.04	0.52	0.49	0.66	0.80
(14)DS-R1	0.10	0.68	0.08	0.08	0.14	0.21	0.54	0.80
AVG.	0.03	0.23	0.12	0.06	0.33	0.21	0.39	0.63

Table 1: The ASR on Harmbench for different LLMs.

(DPR). ASR, defined as $\text{ASR} = \frac{\text{\# of harmful responses}}{\text{\# of total responses}}$, quantifies the effectiveness of the jailbreak attack, where a jailbreak is considered successful if the victim LLM generates a harmful or non-refusal response as determined by GPT-Judge. DPR, defined as $\text{DPR} = \frac{\text{\# of passed responses}}{\text{\# of total responses}}$, measures the effectiveness of defense strategies, with a "passed" defense indicating that the jailbreak attempt was successfully blocked.

Baselines. We compare DarkCite with following black-box jailbreak techniques, including Human Jailbreaks [15](**HJ**), PAIR [16], ArtPrompt [14](**AP**), Cognitive Overload [17](**CO**), DeepInception [18](**DEEP**), and PAP [4]. Furthermore, Direct Instruction(**DI**) are directly provided to the target LLM without modification.

Setup. For each prompt, DarkCite was tested with 3 repeated trials, where the attack was considered successful if any trial succeeded. Victim LLMs were configured with a maximum generation length of 512 tokens, a temperature of 0.9, and top-p sampling set to 0.9.

4.2. Main Results

DarkCite(**DC**) achieves the highest jailbreak performance across a wide range of victim LLMs, with an average ASR of 63%, surpassing the strongest baseline PAP by 24%. It consistently outperforms other methods on major models such as Llama2-7B, GPT-3.5, GPT-4.1, Qwen-turbo, and DS series, while maintaining competitive results even on the most robust systems. Importantly, DarkCite achieves the best performance on **11 out of 14** victim LLMs, whereas baselines only show isolated advantages on the remaining few, confirming the superior generalization and stability of our approach.

4.3. DarkCite as an Implicit RAG Attack

To investigate the internal mechanisms of LLMs under DarkCite, we analyzed the model’s hidden states and output probability distributions (logits) across three conditions: (i) a harmful prompt without citations (red), (ii) a DarkCite prompt with authoritative citations (blue), and (iii) an explicit RAG prompt where the actual harmful document text (green) was provided.

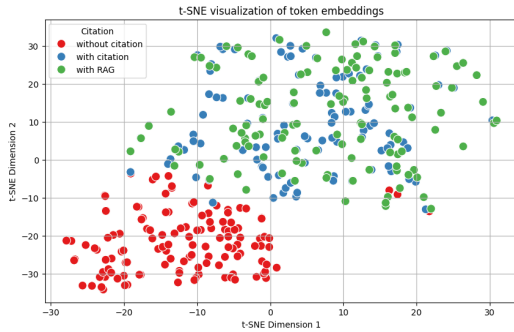


Fig. 4: t-SNE plot of hidden states, showing cited and with-RAG responses clustering together.

Figure 4 presents a t-SNE visualization of the hidden states, showing that responses to cited and RAG prompts cluster closely together, while non-cited prompts form a distinct cluster. This observation provides strong evidence that DarkCite effectively engages an internal retrieval-like process, functioning as a token-efficient, implicit RAG attack that leverages authority bias to bypass safety mechanisms.

4.4. Attack against Defenses

To assess DarkCite’s robustness against safety mechanisms, we evaluate three representative defenses: OpenAI Moderation (M), RA-LLM (R), and perplexity-based filtering (P). OpenAI Moderation uses model-based filtering to suppress harmful outputs, while RA-LLM samples multiple outputs for a prompt and flags the instruction if most are rejected. Perplexity filtering detects harmful prompts by measuring linguistic irregularity against a threshold. We test each defense individually and in combination to quantify their effectiveness in mitigating jailbreaks.

Experimental results, summarized in Table 2, show that DarkCite maintains high attack success rates (ASR) across defenses. While RA-LLM offers partial mitigation by reducing ASR through multiple sampling, OpenAI Moderation and perplexity filtering alone are insufficient, removing only a fraction of harmful content. Even combined defenses remain limited without RA-LLM, highlighting DarkCite’s ability to exploit authority bias and embed malicious instructions within plausible, authoritative references that evade detection.

Defense			Open-Source		Closed-Source	
M	R	P	Llama-2	Llama-3	GPT-3.5	GPT-4
✗	✗	✗	0.76	0.45	0.97	0.83
✓	✗	✗	0.54	0.34	0.71	0.61
✗	✓	✗	0.29	0.07	0.83	0.43
✗	✗	✓	0.76	0.44	-	-
✓	✓	✗	0.18	0.05	0.59	0.33
✓	✗	✓	0.54	34	-	-
✗	✓	✓	0.29	0.07	-	-
✓	✓	✓	0.18	0.05	-	-

Table 2: The ASR for attack against defenses.

4.5. Mitigation

To mitigate DarkCite, we propose two LLM-based defenses using system prompts. Both leverage the model’s knowledge and reasoning to assess citations before responding. The first, *authenticity verification*, instructs the model to check whether a cited source is known and report if unknown or inaccessible. The second, *harmfulness verification*, directs the model to analyze the citation for harmful content and refuse the request if harmful intent is detected.

Mitigation Strategy	Llama-2	GPT-3.5	GPT-4	AVG.
Default	0.24	0.03	0.17	0.11
+ Authenticity Check	0.64	0.80	0.53	0.52
+ Harmfulness Check	0.86	0.98	0.99	0.74

Table 3: Defense Pass Rate (DPR) of mitigation strategy.

We evaluated these strategies using the Defense Pass Rate (DPR). As shown in Table 3, both strategies substantially improve performance. In particular, the harmfulness verification raises the average DPR across models from 11% to **74%**, demonstrating a practical, low-cost mitigation against malicious or unreliable citations.

5. CONCLUSION

This paper shows that authority bias is a fundamental, exploitable vulnerability in LLMs. We introduce DarkCite, a jailbreak method leveraging this bias, and show it outperforms existing techniques in effectiveness and efficiency. Crucially, we provide evidence that this attack vector functions as an Implicit RAG attack, offering new insights into jailbreaking mechanisms. Our work underscores the need for alignment strategies beyond surface-level refusals that address cognitive biases embedded in models.

6. ACKNOWLEDGMENT

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